OLA RIDES

Data Visualization & Analytics — Capstone 2

Final Project Report

Project Title: OLA Rides — Cancellation, Revenue & Operations Intelligence Platform

Sector: Ride-Hailing / Urban Mobility

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GitHub URL: [*https://github.com/suhanigupta786/SectionE_G9_OlaDataset/tree/main*

Tableau URL:*https://public.tableau.com/app/profile/suhani.gupta2725/viz/OLABusinessDashboardAnalysis/Dashboard1?publish=yes*

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1. Executive Summary

OLA Rides processed 49,866 ride requests during January 2024, generating ₹34.18 million in revenue while losing ₹16.85 million due to a 33.01% cancellation rate. This report presents a multi-dimensional data analytics study of OLA's ride operations, covering business health, customer experience, fleet operations, and financial impact. The analysis was conducted using a Python-based ETL and EDA pipeline, with results visualised across four Tableau dashboards designed for CEO, operations, customer-experience, and finance stakeholders respectively.

Problem

One in three rides ends without completion, costing OLA over ₹16.8 million in a single month. Driver cancellations alone account for ₹2.02 million of lost revenue, and a recovery potential of ₹2.53 million exists but remains untapped. The business lacks a consolidated, decision-ready view of where, when, and why cancellations occur.

Approach

Raw booking data was cleaned and transformed in Python (pandas, numpy). KPIs were engineered and validated. Four role-specific Tableau dashboards were built: CEO Business Health, Customer Experience, Operations, and Financial Impact.

Key Insights (Top 4)

33% cancellation rate — one-third of all demand is lost; the recoverable portion alone represents ₹2.53M.

Driver cancellations (19.79%) outweigh customer cancellations (7.58%), pointing to a supply-side problem.

Late Night sees peak successful completions (1,372 trips) but Area-34 records 28% cancellation risk during Early Morning.

Prime Sedan leads all vehicle categories in revenue (₹1.02M) yet also carries the second-highest driver cancellation rate (19.66%).

Key Recommendations (Top 3)

Launch a driver incentive programme targeting high-cancellation areas (Area-34, 36, 37) during Early Morning and Morning slots.

Investigate and fix the 'Wrong Address' issue — the #1 customer cancellation reason — through better address validation at booking.

Deploy dynamic supply rebalancing to shift fleet toward high-demand areas (Area-39, 4, 8) during peak hours.

2. Sector & Business Context

India's ride-hailing sector is among the fastest-growing in the world, with Ola Cabs serving over 250 cities and competing with Uber, Rapido, and Namma Yatri. The sector faces persistent challenges around driver retention, demand-supply mismatches in urban peripheries, and customer trust erosion from frequent cancellations.

The primary decision-maker for this analysis is the OLA CEO and the VP of Operations. A granular understanding of cancellation drivers, revenue leakage, and fleet utilisation is directly actionable at the strategic and city-operations level.

Solving the cancellation problem has compounding value: every 1% reduction in cancellation rate on 49,866 monthly requests translates to approximately 499 additional completed rides and roughly ₹337,000 in incremental monthly revenue (at ₹676.7 average revenue per ride).

3. Problem Statement & Objectives

Formal Problem Definition

OLA's January 2024 booking data reveals that 33.01% of ride requests did not result in a completed trip. The business is unable to quantify, attribute, or act on this loss in real time. The project asks: Which combination of location, time, and vehicle type drives the highest cancellation risk, and what interventions can reduce revenue loss?

Scope

In Scope: January 2024 booking data; all vehicle types; all 50 areas; cancellation analysis, revenue analysis, fleet utilisation, customer experience.

Out of Scope: Driver-level profiling, customer identity data, real-time streaming, pricing strategy.

Success Criteria

All four Tableau dashboards are live and filterable by vehicle type, location, time of day, and hour.
At least 8 distinct, decision-ready insights are documented.
Each recommendation maps to a quantified business impact estimate.
Python notebooks for cleaning, EDA, and statistical analysis are committed to GitHub.

4. Data Description

Dataset Source: OLA Rides Booking Dataset (Kaggle / provided dataset)

Time Period: January 2024 (Dec 30, 2023 – Jan 31, 2024)

Rows: 49,866 booking records

Columns: Approx. 15–20 fields including booking ID, vehicle type, pickup location, booking status, ride distance, driver/customer ratings, timestamps, and revenue fields

Key Fields

Field	Description
Booking ID	Unique identifier per ride request
Vehicle Type	Auto, Bike, eBike, Mini, Prime Plus, Prime Sedan, Prime SUV
Pickup Location	Area code (Area-1 to Area-50)
Booking Status	Success, Cancelled by Customer, Cancelled by Driver, Incomplete
Ride Distance (km)	Distance bracket: 0, 5, 10, 15, 20, 25, 30, 35, 40, 45 km
Revenue	Revenue earned per completed trip (₹)
Customer Rating	1–5 post-trip rating by customer
Driver Rating	1–5 post-trip rating by driver
Date / Hour	Date and hour of booking creation
Cancel Reason	Reason code (customer or driver side)
VTAT / CTAT	Vehicle/Customer Time-At-Target — arrival time metrics

Data Limitations

Single-month snapshot — seasonal generalisation is limited.
Area codes are anonymised; geographic mapping is unavailable.
Driver-level and customer-level IDs are not present, preventing cohort analysis.
Revenue figures for cancelled rides reflect estimated lost revenue, not an audited figure.

5. Data Cleaning & ETL Pipeline

All primary data preparation was performed in Python using pandas and numpy. The notebook is committed at notebooks/02_cleaning.ipynb in the project GitHub repository.

Cleaning Steps

Missing Values: Checked all columns for nulls. Revenue fields for cancelled rides were null — imputed using the average revenue per ride (₹676.7) for 'Revenue Lost' estimation. Rating fields with nulls (cancelled rides have no ratings) were excluded from rating aggregations.

Data Types: Date fields parsed to datetime. Hour extracted as integer. Booking status and vehicle type standardised to title case. Ride distance converted from string to integer (km).

Outlier Treatment: Revenue outliers inspected using IQR method. No records were dropped; extreme values were flagged in the EDA notebook.

Feature Engineering: Created: Is_Successful (boolean), Time_Of_Day (Early Morning / Morning / Afternoon / Evening / Night / Late Night based on hour), Revenue_Lost (estimated for cancelled bookings), Cancel_Flag (binary), and VTAT/CTAT numeric fields.

Standardisation: Pickup location codes normalised. Cancellation reason text cleaned and mapped to 5 standard categories.

Assumptions: Rides with status 'Incomplete' are treated as partial failures and excluded from completed-trip revenue. Revenue Lost for driver cancellations is calculated at the average fare rate.

6. KPI & Metric Framework

KPI	Formula	Jan 2024 Value	Business Relevance
Total Ride Requests	COUNT(Booking_ID)	49,866	Measures total demand volume.
Completion Rate	Successful / Total Requests × 100	66.99%	Primary efficiency KPI; target >75%.
Cancellation Rate	(Cancelled + Incomplete) / Total × 100	33.01%	Core problem metric; industry benchmark ~15%.
Driver Cancel Rate	Driver Cancellations / Total × 100	19.79%	Supply-side reliability indicator.
Customer Cancel Rate	Customer Cancellations / Total × 100	7.58%	Demand-side satisfaction proxy.
Revenue Earned	SUM(Revenue) for Successful rides	₹34.18M	Actual top-line revenue.
Revenue Lost	SUM(Estimated Revenue) for Cancelled rides	₹16.85M	Opportunity cost of cancellations.
Recovery Potential	Revenue Lost attributable to Driver Cancellations only	₹2.53M	Directly actionable revenue recovery.
Avg Revenue per Ride	Revenue Earned / Completed Trips	₹676.7	Unit economics benchmark.
Avg Customer Wait Time	MEAN(CTAT) in minutes	15.57 min	Service quality metric.
Avg Customer Rating	MEAN(Customer_Rating) for completed rides	4.002	Satisfaction index.
Fleet Completion Efficiency	Successful Rides / Total Dispatched Vehicles × 100	66.45%	Fleet productivity measure.

7. Exploratory Data Analysis (EDA)

EDA was performed in notebooks/03_eda.ipynb. Key findings are summarised below by analysis type.

7.1 Trend Analysis — Daily Ride Volume

Daily ride volume stayed consistently between 1,550 and 1,650 bookings throughout January 2024, with no significant spikes or dips. This stability suggests OLA's demand base is relatively inelastic to day-of-week effects in January, but also that cancellation problems are structural rather than event-driven.

7.2 Hourly Demand Trend

Hourly ride demand is broadly flat at approximately 2,000 bookings per hour, with a subtle uptick around hours 11–13. There is no dramatic rush-hour spike, which may indicate either a data aggregation artefact or that OLA's customer base in this dataset spans non-peak commuter segments. Late Night (11 PM–5 AM) is surprisingly the highest-completing time slot (1,372 successful trips), suggesting demand is genuine and under-supplied at night.

7.3 Segment Analysis — Vehicle Type

All seven vehicle types (Auto, Bike, eBike, Mini, Prime Plus, Prime Sedan, Prime SUV) show nearly identical booking volumes (6,984–7,230 rides each), indicating no structural preference skew in the dataset. Revenue is highest for Prime Sedan (₹1.02M) due to higher per-ride fares. Driver cancellation rates are highest for Prime Plus (21.63%) and lowest for Mini (18.24%).

7.4 Booking Status Distribution

Of total bookings: 66.99% completed successfully, 19.22% were cancelled by drivers, 7.58% by customers, and 6.21% were incomplete. The driver cancellation share (19.22%) is more than twice the customer cancellation share, indicating the primary problem is supply-side reliability, not customer behaviour.

7.5 Location Analysis — Top Demand & Revenue Loss Areas

Area-39 has the highest demand (1,097 bookings), followed by Area-4 (1,054) and Area-8 (1,048). However, revenue loss is highest in Area-34 (₹367,379), Area-36 (₹362,262), and Area-37 (₹346,912). This mismatch indicates that high-demand areas are not necessarily the worst-served: the revenue-loss problem is concentrated in specific mid-tier demand zones.

7.6 Customer Experience Analysis

Average customer and driver ratings are nearly identical (4.002 and 4.003), suggesting no systematic bias in rating behaviour. Wait times are uniform across vehicle types (~15.5 min), pointing to a systemic dispatch issue rather than category-specific inefficiency. The most common customer cancellation reason is 'Wrong Address' (>700 occurrences), followed by 'Driver is not moving towards pickup' and 'Driver asked to cancel'.

7.7 Cancellation Risk Heatmap

The heatmap of cancellation rates by area and time of day reveals that Area-34 in Early Morning (28.0%) and Morning (26.9%) carries the highest cancellation risk. Area-1 in Evening (25.3%) and Area-10 and Area-12 in Late Night (24.2%, 24.3%) are also elevated-risk combinations. These are directly actionable hotspots for driver incentive targeting.

8. Statistical Analysis

Statistical analysis was performed in notebooks/04_statistical_analysis.ipynb.

8.1 Cancellation Rate Comparison (Chi-Square Test)

A chi-square test of independence was conducted to determine whether cancellation rates differ significantly across vehicle types. The null hypothesis (cancellation rate is independent of vehicle type) was [rejected / not rejected — fill in p-value and χ^2 statistic from notebook]. This supports targeting vehicle-specific retention interventions.

8.2 Revenue Loss Segmentation

K-means clustering on (area, revenue_lost, cancellation_rate) identified three area segments: High-Loss High-Risk (Area-34, 36, 37, 13), Moderate-Loss (Area-5, 1, 10, 19), and Low-Loss Low-Risk (remaining areas). The high-loss cluster accounts for approximately 42% of total driver-cancelled revenue despite representing only 20% of areas.

8.3 Correlation Analysis

Pearson correlation between average VTAT (vehicle arrival time) and cancellation rate at the area level shows a positive relationship: areas with longer average arrival times have higher driver cancellation rates. This supports the hypothesis that longer distances to pickup disincentivise drivers. [Insert r-value from notebook].

8.4 Forecasting (Optional Extension)

A 7-day rolling average was applied to daily ride volume to smooth noise and project forward. Given the stable trend in January, a simple linear projection [or ARIMA if implemented] suggests February demand will remain in the 1,550–1,650 daily booking range absent any supply-side changes. [Update with actual model output from notebook].

9. Tableau Dashboard Design

Four Tableau dashboards were designed, each serving a distinct stakeholder. All dashboards share a consistent dark theme (black background, lime-yellow accent, OLA branding) and support cross-filtering via shared filters: Vehicle Type, Pickup Location, Time of Day, and Hour.

Dashboard 1: CEO — Business Health

Objective: Gives the CEO a single-screen view of total demand, revenue earned vs lost, daily volume trends, vehicle category performance, and booking status breakdown.

Visuals Included:

- Total Requests, Completion Rate, Cancellation Rate, Revenue Earned, Revenue Lost, Recovery Potential (KPI cards)
- Revenue Earned vs Revenue Lost (grouped bar by vehicle type)
- Top 5 Pickup Locations by Demand (horizontal bar)
- Daily Ride Volume Trend (line chart, Dec 30 – Jan 29)
- Demand Trend by Hour (line chart)
- Vehicle Category Chart (bar, sorted by volume)
- Booking Status Distribution (donut chart)

Dashboard 2: Customer Experience

Objective: Helps the Customer Experience team diagnose wait-time problems, cancellation reasons, and location-level availability issues.

Visuals Included:

- Avg Customer Rating, Avg Driver Rating, Avg Wait Time, Customer Cancel Rate, Most Common Reason (KPI cards)
- Customer Wait Time by Vehicle Type (bar)
- Customer vs Driver Ratings Comparison (grouped bar by vehicle type)
- Preferred Vehicle Type — successful bookings (bar)
- Location Ride Availability Issues (bar by area)
- Ride Availability by Time of Day (bar)
- Customer Cancellation Reasons (horizontal bar)

Dashboard 3: Operations

Objective: Enables the Fleet Operations team to monitor driver cancellations, fleet utilisation, VTAT performance, and cancellation risk by area and time.

Visuals Included:

- Driver Cancellation Rate, Avg Vehicle Arrival Time, Fleet Completion Efficiency, Total Completed Trips (KPI cards)
- Driver Cancellation by Vehicle Type (bar with % labels)
- Fleet Utilisation by Time Slot (bar, 6 time slots)

Average VTAT by Area — Top 10 (horizontal bar)
Cancellation Risk Heatmap (area × time of day, colour-coded green–red)

Dashboard 4: Financial Impact

Objective: Supports the CFO and revenue team in understanding where and how revenue is being lost, which locations are most efficient, and how vehicle type drives revenue.

Visuals Included:

Revenue Lost (Driver), Revenue Recovery Reduction, Avg Revenue per Ride, Revenue Loss Rate (KPI cards)
Revenue Lost by Top 10 Areas (horizontal bar)
Location-wise Completion Efficiency (horizontal bar)
Revenue by Vehicle Type (bar with ₹ labels)
Distance per Trip Distribution (histogram by km bracket)

Tableau Public URL:
https://public.tableau.com/app/profile/suhani.gupta2725/viz/OLABusinessDashboardAnalysis/Dashboard1?publish=yes

Dashboard screenshots are included in Appendix A and in tableau/screenshots/ in the GitHub repository.

10. Insights Summary

The following 10 insights are written in decision language — each tells the reader what to think or act upon.

INS-01 A 33% cancellation rate is nearly double the acceptable industry threshold (~15%), meaning OLA is operating at roughly two-thirds capacity and leaving a structural revenue gap every month.

INS-02 Driver cancellations (19.22%) are the dominant failure mode — more than twice customer cancellations. Fixing supply-side reliability will have far greater impact than improving customer retention alone.

INS-03 Area-34, Area-36, and Area-37 collectively lose over ₹1.07M per month and are the highest-priority intervention zones for driver incentive programmes.

INS-04 'Wrong Address' is the #1 customer cancellation reason (>700 occurrences), indicating a solvable product problem: improved address autocomplete or pre-booking confirmation could eliminate a significant share of these cancellations.

INS-05 Late Night (11 PM–5 AM) is the highest-completing time slot (1,372 trips) but also underserved by fleet supply — there is an opportunity to increase revenue by deploying more drivers during late-night hours.

INS-06 Prime Plus vehicles carry the highest driver cancellation rate (21.63%), while Mini carries the lowest (18.24%). Segment-specific retention strategies are warranted; blanket incentives are inefficient.

INS-07 All vehicle types show nearly identical average wait times (~15.5 min) — this uniformity across categories suggests the wait-time problem is not category-specific but stems from low fleet density overall.

INS-08 Fleet completion efficiency is 66.45% — for every 100 vehicles dispatched, 34 fail to complete a ride. Closing this gap by even 5 percentage points would add ~2,500 completed rides per month.

INS-09 Area-50 leads location-wise completion efficiency (68.98%), while Area-36 is the least efficient (63.28%). Studying what makes Area-50 more successful could yield transferable operational best practices.

INS-10 Revenue recovery potential of ₹2.53M exists specifically from driver-cancelled rides — this is directly targetable through driver behaviour intervention without acquiring new customers.

11. Recommendations

REC-01 — Targeted Driver Retention Incentive (High Priority)

Linked Insight: INS-02, INS-03

Recommendation: Launch a time-location bonus scheme for drivers completing rides in Area-34, 36, 37, and 13 during Early Morning and Morning slots. Offer ₹50–100 per completed ride above a threshold.

Expected Impact: Reducing driver cancellation rate from 19.79% to 15% would recover ~2,380 rides/month × ₹676.7 = ~₹1.61M additional monthly revenue.

Feasibility: Implementable within the existing driver app infrastructure. Requires ₹50–100 incentive budget vs. ₹1.61M potential uplift — strong ROI.

REC-02 — Address Validation Product Fix (High Priority)

Linked Insight: INS-04

Recommendation: Integrate enhanced address autocomplete and a pre-booking address confirmation step (map pin confirmation) to eliminate 'Wrong Address' cancellations.

Expected Impact: Eliminating 50% of 'Wrong Address' cancellations (~350 rides/month) = ~₹237K additional monthly revenue.

Feasibility: Product engineering task; estimated 2–4 sprint cycles. Low cost, permanent fix.

REC-03 — Late Night Supply Expansion (Medium Priority)

Linked Insight: INS-05

Recommendation: Deploy a 'Night Owl' driver incentive for 11 PM–5 AM shifts in high-demand areas (Area-39, 4, 8). Target 15% increase in available night-time supply.

Expected Impact: If 15% more late-night demand is served (estimated 200 additional rides/month at ₹676.7) = ~₹135K/month.

Feasibility: Requires driver communication via app notification. Low implementation cost.

REC-04 — Vehicle-Segment-Specific Cancellation Programme (Medium Priority)

Linked Insight: INS-06

Recommendation: Design separate driver retention strategies for Prime Plus (highest cancel rate) vs. Mini (lowest). For Prime Plus, investigate whether longer trip distances cause cancellations and offer mileage bonuses.

Expected Impact: Even a 2% reduction in Prime Plus cancellation rate (~146 additional rides) adds ~₹99K/month.

Feasibility: Data available to segment; requires two campaign variants.

REC-05 — Area-50 Best-Practice Replication (Lower Priority)

Linked Insight: INS-09

Recommendation: Commission an operational audit of Area-50 (highest completion efficiency at 68.98%) to identify driver behaviour, demand patterns, or dispatch rules that can be replicated in low-efficiency areas like Area-36.

Expected Impact: If Area-36 completion efficiency improves from 63.28% to 68%, it adds ~₹47K/month in that area alone.

Feasibility: Operational research effort, no technology investment required.

12. Impact Estimation

The table below summarises estimated monthly revenue impact from implementing the recommended actions. All figures are conservative estimates based on partial adoption scenarios.

Action	Revenue Uplift	Estimated Cost	Net Impact
REC-01: Driver Incentive Programme	~₹1,610,000	~₹500,000 incentive cost	Net ~₹1.11M/month
REC-02: Address Validation Fix	~₹237,000	One-time engineering effort	~₹237K/month ongoing
REC-03: Late Night Supply Expansion	~₹135,000	~₹30,000 incentive cost	Net ~₹105K/month
REC-04: Vehicle-Segment Programme	~₹99,000	~₹25,000 campaign cost	Net ~₹74K/month
REC-05: Area-50 Best-Practice Replication	~₹47,000	Operational audit cost	~₹47K/month per area
TOTAL (all 5 recs, conservative)	~₹2,128,000	~₹555,000 total cost	Net ~₹1.57M/month

Annualised, the five recommendations combined could recover approximately ₹18.8M per year at a net cost of ₹6.66M — a 2.8× return on intervention investment. The business case for acting immediately is strong: every month of delay costs roughly ₹1.57M in recoverable revenue.

13. Limitations

Single-month data: January 2024 findings may not reflect seasonal demand patterns. Festive months or monsoon season could substantially alter cancellation behaviour.
Anonymised locations: Area codes prevent geographic clustering or distance-to-city-centre analysis, limiting spatial interpretation.
No driver-level or customer-level IDs: Repeat cancellation behaviour by specific drivers or customers cannot be tracked.
Revenue imputation: Revenue Lost for cancelled rides is estimated at average fare; actual lost revenue may differ based on trip distance, surge pricing, and vehicle type.
No competitive data: OLA's cancellation rate cannot be benchmarked against Uber or Rapido for January 2024.
Causal inference limits: The heatmap and correlation analysis identify associations, not causal mechanisms. Driver incentive design should be piloted and tested before full rollout.

14. Future Scope

Multi-month analysis (Jan–Dec 2024) to detect seasonality in cancellations and revenue.
Driver-level profiling: if driver IDs become available, build a driver reliability score to proactively flag high-cancellation drivers.
Predictive cancellation model: logistic regression or gradient boosting model to predict, at booking time, the probability of cancellation based on area, time, and vehicle type.
Real-time dashboard: Connect Tableau (or a Streamlit app) to a live data feed for operational monitoring.
Geo-spatial analysis: Map Area codes to actual city zones using OLA's internal geography data to enable district-level analysis.
Customer LTV analysis: Combine cancellation history with repeat booking frequency to identify high-value customers at risk of churn.

15. Conclusion

OLA's January 2024 data reveals a business operating under significant self-imposed drag: a 33% cancellation rate — driven primarily by driver-side failures in specific high-loss areas and time slots — is costing the platform ₹16.85 million per month in direct revenue loss. Through Python-based ETL and EDA and four role-specific Tableau dashboards, this project has quantified the problem, identified its hotspots (Area-34, 36, 37), diagnosed its root causes (driver incentive gaps, wrong-address bookings, late-night supply shortage), and proposed five prioritised, costed recommendations with a combined net monthly recovery potential of ₹1.57 million. The highest-priority action — a targeted driver retention incentive in three areas — alone offers a 2.2× return on investment within the first month of deployment. The data infrastructure and dashboards built in this project provide OLA's leadership team with the decision-ready visibility needed to move from reactive problem-solving to proactive, location-time-specific fleet management.

17. APPENDIX

Data Dictionary

A detailed data dictionary with column definitions is available in the project repository:
https://github.com/suhanigupta786/SectionE_G9_OlaDataset/tree/main

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Additional Analysis

Additional exploratory analysis and supporting visualizations are included in the EDA notebook:

https://github.com/suhanigupta786/SectionE_G9_OlaDataset/blob/main/notebooks/03_eda.ipynb

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Data Cleaning & Transformation

All data cleaning, preprocessing, and feature engineering steps are documented in:
https://github.com/suhanigupta786/SectionE_G9_OlaDataset/blob/main/notebooks/02_cleaning_final.ipynb

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Dashboard Access

The interactive Tableau dashboards can be accessed here:

https://public.tableau.com/app/profile/suhani.gupta2725/viz/OLABusinessDashboardAnalysis/Dashboard1?publish=yes

Contribution Matrix

Fill in team member names and mark contributions with ✓ or percentage.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
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Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.

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